

Amplitude–Slope Compensation in Feedback Transseries

Sharp regularity, negative-ray summation,
and damping-induced condensation

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Abstract

Consider the formal equation

$$\widehat{U}(q) = \sum_{j \geq 1} c_j q^j \exp(\lambda_j \widehat{U}(q)), \quad c_1 = 1, \quad 0 < c_j \leq 1, \quad \lambda_j \geq 0,$$

and let $d_j = \log(1/c_j)$. We prove that its solution converges precisely when $\lambda_j = O(j + d_j)$. For every $s > 0$, the solution and its compositional inverse are Gevrey– s precisely when

$$\lambda_j^{1/(s+1)} = O(j \log(j+1) + d_j).$$

The same condition characterizes a uniform Gevrey– s expansion of the literal inverse on a closed left half-disc. At $s = 1$ it is also equivalent to fine Borel–Laplace summability of both series in the negative direction; the sums are inverse functions and satisfy the original convergent kernel. Under strong damping, $d_j/(j \log(j+1)) \rightarrow \infty$, the exact factorial-normalized type is

$$T_s(\widehat{U}) = (s+1)^{s+1} \limsup_{j \rightarrow \infty} \frac{\lambda_j}{d_j^{s+1}}.$$

For $d_j \sim bj^\beta$, $\lambda_j \sim aj^p$, $p > \beta > 1$, the type is an actual coefficient-root limit, and the dominant action has size $((p/\beta)n/b)^{1/\beta}$ in probability. These results answer the repository’s weighted-amplitude question for positive, exponentially bounded amplitudes. The package includes exact finite verification, a rational inverse certificate, numerical diagnostics, and further research questions.

Status and scope. These are conventional proofs, not Lean-verified statements. Global publication priority has not been established. The contribution claimed is a precise extension of the inspected ProveIt research, not a solution of an unrelated named conjecture. Fine summability means a fixed-width neighborhood of one Borel ray; angular summability and resurgence are not asserted.

Contents

1	The question addressed and the contribution	4
1.1	Repository provenance	4
1.2	Why amplitudes change the problem	4
1.3	What is new relative to the inspected sources	4
2	Definitions and the main results	5
2.1	Normalization and formal series	5
2.2	Positive amplitudes with an exponential upper bound	6
3	Finite algebra and the necessary estimates	7
3.1	Existence, positivity, and Lagrange coefficients	7
3.2	Necessity of the joint Gevrey budget	7
3.3	Gevrey membership is invariant under formal reversion	7
4	The exact convergence criterion	8
5	The weighted exponential-block inverse	8
5.1	Explicit blocks	8
5.2	A literal inverse for every slope sequence	9
5.3	A finite block certificate	9
6	Sufficiency and uniform Gevrey remainders	9
6.1	A weighted block-moment estimate	10
6.2	Coefficient sufficiency	10
6.3	The same budget controls an actual analytic remainder	11
7	Bessel kernels and fine negative-ray summation	11
7.1	One exact Borel kernel	11
7.2	The all-slope ray transform is not automatically a Borel germ	12
7.3	Amplitude damping gives a complex neighborhood	12
7.4	Growth on a fixed-width tube	13
7.5	Passing from the inverse to the forward series	13
8	Exact forward type in the strong-damping regime	13
8.1	A universal upper-type lemma	13
8.2	A matching one-action lower bound	14
8.3	What the type identity does and does not say	15
9	Dense power costs and a dominant-action limit	15
9.1	Obtaining the lower bound at every degree	15
9.2	The probability space and the length concentration	15
9.3	Total cost is concentrated at the unique variational maximum	16
9.4	A single part carries asymptotically all the cost	16

10 A sharper Borel region under strong damping	17
11 Phase diagrams and explicit examples	17
11.1 Power-law slopes and stretched-exponential amplitudes	17
11.2 Logarithmic boundary factors	18
11.3 Superpolynomial slopes can still be summable	19
12 Why the hypotheses and the direction matter	19
12.1 Exponential amplitude control is a substantive assumption	19
12.2 Positivity is not just a convenient estimate	19
12.3 Why positive-ray summation fails for divergent positive series	19
12.4 Remaining distinctions	20
13 Exact checks, numerical diagnostics, and reproducibility	20
13.1 Independent finite identities	20
13.2 A rational certificate for a literal inverse value	20
13.3 Coefficient-root diagnostics	21
13.4 Reproduction commands and dependencies	21
14 Formalization targets and proof dependencies	22
14.1 Finite coefficient algebra	22
14.2 Real estimates and the factorial budget	22
14.3 Analytic realization and summation	22
14.4 Sharp asymptotics and probability	22
15 Further research questions	23
15.1 Sharp signed inverse coefficients	23
15.2 A full multiplicative forward asymptotic	23
15.3 Local fluctuations and a conditional ensemble	23
15.4 Lacunary cost spectra and log-periodic behavior	23
15.5 Actual inverse Borel singularities	23
15.6 Angular and generalized moment summation	23
15.7 Amplitudes beyond exponential growth	24
15.8 Complex phases and vector feedback	24
15.9 General actions and missing action one	24
15.10 Certified algorithms with adaptive costs	24
16 Conclusion	24
A Compositional inversion in the fine-summable class	25
A.1 The Borel minor at infinity	25
A.2 An explicit inverse minor	25
A.3 Returning to series tangent to the identity at zero	26
B Notation and the logical dependency map	27

1 The question addressed and the contribution

1.1 Repository provenance

The source snapshot inspected for this article is

ebd8344bca77d8352cd745b7df374618a290a029.

The relevant project is **Analysis/Transseries/**. Its directory README describes the recent research packages as unmerged and not formalized [1]. We use their mathematical questions and methods as starting points, not their presence in a repository as a certificate of correctness.

Two specific source passages motivate the work. The regularity article asks for an amplitude-dependent analogue of its unit-amplitude results [2, lines 1470–1483]. The negative-ray article asks for a sharp joint condition on amplitudes and slopes, including very rapidly decaying amplitudes [3, section “Weighted actions and joint amplitude–slope thresholds”]. The latter source proves a unit-amplitude fine-summation theorem. The present paper treats an arbitrary positive amplitude sequence with an exponential upper bound. It does not change the action exponents from the integers j .

The focused review included the project and document-directory READMEs, the relevant regularity questions, and the negative-ray article’s statements, inverse representation, continuation argument, and open questions. It was not an exhaustive audit of the canonical volume or of every concurrent research arrival. Reproducible source pointers are recorded in `SOURCE_NOTES.md`.

1.2 Why amplitudes change the problem

For unit amplitudes, a large slope λ_j can be used repeatedly through the exponential expansion without paying an initial suppression cost. A small amplitude c_j changes the balance. Entering action j now costs $d_j = -\log c_j$ in the logarithm of a positive coefficient. The cost can be larger than the combinatorial penalty associated with its action index. It can even compensate for slopes growing faster than any polynomial.

Three separate questions must therefore be distinguished. First, does the formal solution have a nonzero ordinary convergence radius? Second, which factorial growth bounds do its coefficients satisfy? Third, does an analytic realization recover those coefficients with uniform remainders and a canonical directional summation rule? Our results address all three, but the analytic remainder statement for arbitrary s is made for the inverse. Fine summability of the forward series follows at $s = 1$ from a separate nonlinear argument.

1.3 What is new relative to the inspected sources

The principal additions are the joint necessary-and-sufficient criteria, the weighted uniform remainder theorem, and the exact type identity in the strong-damping regime. The stretched-exponential specialization exhibits a dominant-action scale $n^{1/\beta}$ and a nonzero critical factorial type that differ from the unit-amplitude behavior. No monotonicity or regular variation is needed for the joint criteria or for the strong-damping *limsup* identity.

The algebraic tools are classical. Lagrange inversion is used in its standard coefficient form [4]. The Borel kernels are modified Bessel functions, with an elementary integral representation [6]. The distinction between fine and angular summability, and the convolution approach to nonlinear operations, are standard [5]. The new input is the quantitative use of the amplitude cost throughout those constructions. We prove the specialized estimates needed here rather than appealing to a generic analytic implicit-function theorem for a kernel that may not be analytic at the origin.

2 Definitions and the main results

2.1 Normalization and formal series

Throughout the main proofs assume

$$c_1 = 1, \quad 0 < c_j \leq 1, \quad \lambda_j \geq 0, \quad d_j = -\log c_j \geq 0. \quad (2.1)$$

All individual slopes are finite. Define

$$\Phi(q, u) = \sum_{j \geq 1} c_j q^j e^{\lambda_j u}, \quad \hat{U} = \Phi(q, \hat{U}), \quad \hat{U}(q) = \sum_{n \geq 1} u_n q^n, \quad \hat{Q} = \hat{U}^{\langle -1 \rangle} = \sum_{n \geq 1} v_n u^n. \quad (2.2)$$

Here u is the argument of $\hat{Q}$, not a coefficient. Since $u_1 = 1$, the formal compositional inverse exists. For any fixed degree, every sum used in the formal equation is finite.

Definition 2.1 (Gevrey order and type). For $s > 0$, a formal series $f(z) = \sum f_n z^n$ belongs to $\mathcal{G}_s$ if $|f_n| \leq CH^n (n!)^s$ for some $C, H > 0$. Its exact factorial-normalized type is

$$T_s(f) = \limsup_{n \rightarrow \infty} \left(\frac{|f_n|}{(n!)^s} \right)^{1/n} \in [0, \infty]. \quad (2.3)$$

Thus $f \in \mathcal{G}_s$ if and only if $T_s(f) < \infty$. The convention uses $(n!)^s$, not $\Gamma(1+sn)$; the normalization matters for exact constants.

Set

$$L_j = j \log(j+1), \quad H_j = L_j + d_j. \quad (2.4)$$

A uniform Gevrey expansion of Q on a closed left half-disc means that, for some $r_0, C, A > 0$ and every $N \geq 1$,

$$\left| Q(u) - \sum_{n=1}^{N-1} v_n u^n \right| \leq CA^N (N!)^s |u|^N, \quad \operatorname{Re} u \leq 0, \quad |u| \leq r_0. \quad (2.5)$$

Boundary values on the imaginary axis are furnished by the convergent block construction, not by analytic continuation across that axis.

Theorem 2.2 (Joint convergence and regularity classification). *Under (2.1), the formal solution converges in a neighborhood of zero if and only if*

$$\sup_{j \geq 1} \frac{\lambda_j}{j + d_j} < \infty. \quad (2.6)$$

For every fixed $s > 0$, the following assertions are equivalent:

$$(i) \sup_j \lambda_j^{1/(s+1)} / H_j < \infty;$$

$$(ii) \hat{U} \in \mathcal{G}_s;$$

$$(iii) \hat{Q} \in \mathcal{G}_s;$$

$$(iv) \text{ the literal inverse branch } u = \Phi(Q(u), u) \text{ satisfies (2.5) on a sufficiently small closed left half-disc.}$$

The literal inverse branch exists on the left independently of these growth conditions. No monotonicity hypothesis is required.

Definition 2.3 (Fine Borel summability). Let

$$\mathcal{B}_0 f(\zeta) = \sum_{n \geq 0} \frac{f_n}{n!} \zeta^n, \quad \Omega_\delta^- = \{\zeta : \text{dist}(\zeta, (-\infty, 0]) < \delta\}.$$

A formal series is finely Borel summable in direction π when its Borel germ extends holomorphically to some Ω_δ^- and is bounded there by $Ce^{A|\zeta|}$. Its sum at $u = -r$ is

$$\mathcal{S}_\pi f(-r) = \frac{1}{r} \int_0^\infty e^{-t/r} \mathcal{B}_0 f(-t) dt. \quad (2.7)$$

Theorem 2.4 (Sharp weighted negative-ray summability). *Under (2.1), each of the four conditions of Theorem 2.2 at $s = 1$ is also equivalent to each of*

$$\widehat{U} \text{ is finely summable in direction } \pi, \quad \widehat{Q} \text{ is finely summable in direction } \pi.$$

The sums are compositional inverses near the negative real axis. They satisfy the literal kernel equation on their sufficiently small common left-side domain, with an absolutely convergent sum over j .

Theorem 2.5 (Exact type under strong amplitude damping). *Suppose additionally that*

$$\frac{d_j}{j \log(j+1)} \rightarrow \infty. \quad (2.8)$$

Then for every $s > 0$, with values in $[0, \infty]$,

$$T_s(\widehat{U}) = (s+1)^{s+1} \limsup_{j \rightarrow \infty} \frac{\lambda_j}{d_j^{s+1}}. \quad (2.9)$$

This is a forward-series type theorem. It does not assert equality of forward and inverse exact types.

Theorem 2.6 (Dense power costs: root limit and dominant action). *Assume*

$$d_j \sim bj^\beta, \quad \lambda_j \sim aj^p, \quad a, b > 0, \quad p > \beta > 1. \quad (2.10)$$

Put $r = p/\beta > 1$ and $s = r - 1$. Then

$$\lim_{n \rightarrow \infty} \left(\frac{u_n}{(n!)^s} \right)^{1/n} = a \left(\frac{r}{b} \right)^r. \quad (2.11)$$

Give each ordered composition $(j_1, \dots, j_k)$ of n probability proportional to its nonnegative Lagrange weight in (3.1). If $K = k$, $D = \sum_i d_{j_i}$, and $J_{\max} = \max_i j_i$, then

$$\frac{K}{n} \rightarrow 1, \quad \frac{D}{n} \rightarrow r, \quad \frac{\max_i d_{j_i}}{D} \rightarrow 1, \quad \frac{J_{\max}}{n^{1/\beta}} \rightarrow \left(\frac{r}{b} \right)^{1/\beta} \quad (2.12)$$

in probability. The ratio involving D is defined arbitrarily when $D = 0$, an event whose probability tends to zero.

2.2 Positive amplitudes with an exponential upper bound

The normalization $c_j \leq 1$ does not exclude the usual exponentially bounded case. Suppose $c_1 = 1$, $c_j > 0$, and $\sup_{j \geq 2} c_j^{1/(j-1)} < \infty$. Choose $R \geq 1$ such that $c_j R^{1-j} \leq 1$ and set

$$V(z) = R\widehat{U}(z/R), \quad \tilde{c}_j = c_j R^{1-j}, \quad \tilde{\lambda}_j = \lambda_j/R. \quad (2.13)$$

Then V satisfies the normalized model. With $d_j^+ = \max(0, -\log c_j)$, one has

$$d_j^+ \leq \tilde{d}_j \leq d_j^+ + (j-1) \log R.$$

Consequently the analytic and Gevrey criteria hold for the original model with d_j replaced by d_j^+ . Fine summability and uniform remainder properties survive this positive real rescaling. The strong-damping type formula transforms back to exactly (2.9).

3 Finite algebra and the necessary estimates

3.1 Existence, positivity, and Lagrange coefficients

If $V, W \in q\mathbb{R}[[q]]$ agree through degree $n - 1$, then $\Phi(q, V)$ and $\Phi(q, W)$ agree through degree n . Iterating from zero therefore constructs the unique formal solution. All coefficients are nonnegative, and the coefficient of q is $c_1 = 1$.

Apply Lagrange inversion to $V = t\Phi(q, V)$ in the auxiliary variable t . Substitution $t = 1$ is legitimate coefficientwise because Φ has positive q -order. The coefficient formula is

$$u_n = \sum_{k=1}^n \frac{1}{k!} \sum_{\substack{j_1 + \dots + j_k = n \\ j_i \geq 1}} \left(\prod_{i=1}^k c_{j_i} \right) \left(\sum_{i=1}^k \lambda_{j_i} \right)^{k-1}. \quad (3.1)$$

For $k = 1$ the last factor is interpreted as one, even if its base is zero. Grouping equal parts gives the equivalent finite expression

$$u_n = \sum_{\sum j m_j = n} \frac{\prod_j c_j^{m_j}}{\prod_j m_j!} \left(\sum_j m_j \lambda_j \right)^{\sum m_j - 1}. \quad (3.2)$$

These formulas are identities in finitely many variables at each degree; no analytic convergence is being used.

The term with one part $j \geq 2$ and m parts equal to one proves

$$u_{j+m} \geq c_j \frac{(\lambda_j + m\lambda_1)^m}{m!} \geq e^{-d_j} \frac{\lambda_j^m}{m!}. \quad (3.3)$$

Retaining the amplitude in this inequality is essential.

3.2 Necessity of the joint Gevrey budget

Suppose $u_n \leq CH^n(n!)^s$, enlarging C, H to be at least one. For $j \geq 2$ set $m = \lceil L_j + d_j \rceil$. From (3.3),

$$\lambda_j \leq C^{1/m} H^{1+j/m} e^{d_j/m} (m!)^{1/m} ((j+m)!)^{s/m}. \quad (3.4)$$

The factors $C^{1/m}$, $H^{1+j/m}$, and $e^{d_j/m}$ are bounded. Also $j + m \leq 2m$ and

$$((j+m)!)^{s/m} \leq (j+m)^s \exp\left(s \frac{j}{m} \log(j+m)\right).$$

The last exponential is uniformly bounded. Indeed $m \geq j \log(j+1)$, and $\log(2m)/m$ is decreasing once m is sufficiently large; its maximum in that range is attained, up to finitely many cases, at the lower endpoint. Thus $\lambda_j \leq C_s m^{s+1}$, which is exactly the necessary bound in Theorem 2.2. The finitely many small indices are absorbed into the constant.

3.3 Gevrey membership is invariant under formal reversion

Lemma 3.1. *For $s > 0$ and $f(z) = z + O(z^2)$, one has $f \in \mathcal{G}_s$ if and only if $f^{(-1)} \in \mathcal{G}_s$.*

Proof. Write $f(z) = z(1 + h(z))$. The inverse coefficient is $n^{-1}[z^{n-1}](1 + h(z))^{-n}$. A term with ℓ factors is indexed by positive integers $i_1 + \dots + i_\ell = n - 1$. Their factorials satisfy

$$\prod_{a=1}^{\ell} (i_a + 1)! \leq 2^{\ell-1} (n - \ell + 1)! \leq 2^n n!.$$

To prove this, move units from smaller parts exceeding one to a largest part; the product cannot decrease, and eventually all but one part equal one. Raising the estimate to s , bounding the binomial coefficient from the negative-power expansion by 4^n , and the number of compositions by 2^n , gives $|[z^n]f^{(-1)}| \leq E^n(n!)^s$. The bounds on the input coefficients contribute only another exponential factor. Apply the same argument to the inverse for the reverse implication. $\square$

This deliberately proves only class membership. None of the constants in this lemma are sharp enough to prove an exact type identity.

4 The exact convergence criterion

Suppose first that $\widehat{U}$ converges. Choose $q_0 > 0$ within its radius. Since every coefficient is nonnegative, Tonelli's theorem applied to the formal identity gives an equality of nonnegative sums

$$\widehat{U}(q_0) = \sum_j c_j q_0^j e^{\lambda_j \widehat{U}(q_0)} < \infty.$$

Writing $t = \widehat{U}(q_0) > 0$, each summand is bounded by t , so

$$t\lambda_j \leq d_j + j \log(1/q_0) + \log t.$$

This yields $\lambda_j = O(j + d_j)$.

Conversely, suppose $\lambda_j \leq K(j + d_j)$ with $K > 0$. On a small polydisc with $|u| < 1/(2K)$,

$$c_j |q|^j e^{\lambda_j |u|} \leq (|q|e^{1/2})^j e^{-d_j/2}. \quad (4.1)$$

The right side is summable when $|q| < e^{-1/2}$. Thus Φ is holomorphic in a neighborhood of $(0, 0)$. The analytic implicit-function theorem applies to $u - \Phi(q, u)$ because its u -derivative at the origin is one. Its Taylor series must be the unique formal solution.

This establishes (2.6). It also shows why the $j \log j$ term must not be inserted into the convergence criterion. For example, $c_j = 1$ and $\lambda_j = j \log(j + 1)$ give a divergent solution that belongs to every positive Gevrey class. Ordinary convergence is a strictly different endpoint problem.

5 The weighted exponential-block inverse

5.1 Explicit blocks

For fixed u , first invert $q \mapsto \Phi(q, u)$ with an independent target variable w . Lagrange inversion and a negative-binomial expansion give, after setting $w = u$,

$$Q(u) = \sum_{k \geq 1} u^k \sum_{\substack{\mathbf{m}=(m_1, \dots, m_\ell) \\ m_i \geq 1, \sum_i m_i = k-1}} \omega_{k,\ell} e^{-D_{\mathbf{m}} b_{k,\mathbf{m}} u}, \quad (5.1)$$

where the empty composition is used when $k = 1$, and

$$\omega_{k,\ell} = \frac{(-1)^\ell}{k} \binom{k + \ell - 1}{\ell}, \quad (5.2)$$

$$D_{\mathbf{m}} = \sum_i d_{m_i+1}, \quad (5.3)$$

$$b_{k,\mathbf{m}} = \sum_i \lambda_{m_i+1} - (k + \ell)\lambda_1. \quad (5.4)$$

In particular, the first block is $ue^{-\lambda_1 u}$. Each block starts at ordinary degree k , so (5.1) also makes sense formally. It is exactly $\hat{Q}$, by uniqueness of formal compositional inversion.

Let $C_k = \sum_{\mathbf{m}} |\omega_{k,\ell}|$. For $k \geq 2$,

$$C_k = \frac{1}{k} \sum_{\ell=1}^{k-1} \binom{k+\ell-1}{\ell} \binom{k-2}{\ell-1} \leq 8^k, \quad (5.5)$$

and $C_1 = 1$. The last bound follows directly from $\binom{k+\ell-1}{\ell} \leq 4^k$ and the total number 2^{k-2} of compositions of $k-1$. We use this simple bound rather than an optimized Schröder-number estimate.

5.2 A literal inverse for every slope sequence

Every frequency satisfies

$$b_{k,\mathbf{m}} \geq -2k\lambda_1. \quad (5.6)$$

Consequently the absolute value of block k on $\operatorname{Re} u \leq 0$ is at most

$$(8|u|e^{2\lambda_1|u|})^k.$$

Choose $r_0 > 0$ such that $8r_0e^{2\lambda_1 r_0} < 1/3$. The block series converges uniformly on the closed left half-disc of radius r_0 , and normally in its interior. The same calculation with w in place of the factor u^k proves a uniform convergence radius for the inverse in the independent variable w . Since it agrees near $w = 0$ with the analytic inverse of $q \mapsto \Phi(q, u)$, substitution $w = u$ is valid there. The bound also keeps $|Q(u)| < 1/2$, where the literal kernel converges absolutely.

We have therefore constructed, with no slope-growth assumption,

$$u = \sum_{j \geq 1} c_j Q(u)^j e^{\lambda_j u}, \quad Q(u) = u + O(u^2). \quad (5.7)$$

The latter estimate is uniform on the half-disc. On every smaller closed sector strictly inside the left half-plane, Cauchy's estimate gives $Q'(u) = 1 + O(u)$, so the analytic forward inverse exists near the negative axis. Uniform Gevrey control is an additional assertion, proved next.

5.3 A finite block certificate

Let $Q_{\leq K}$ denote the first K blocks. For $u = -r$ with $r \leq r_0$, put $t = 8re^{2\lambda_1 r} < 1$. Then

$$|Q(-r) - Q_{\leq K}(-r)| \leq \frac{t^{K+1}}{1-t}. \quad (5.8)$$

This certificate does not use Gevrey regularity, and it remains valid for slopes too large for any finite Gevrey class. Only finitely many amplitudes and slopes are needed to evaluate the first K blocks.

6 Sufficiency and uniform Gevrey remainders

Fix $s > 0$ and write $r = s + 1 > 1$. Assume $\lambda_j^{1/r} \leq K(L_j + d_j)$ for all j . Constants in this section may depend on the model and on s , but never on a coefficient order or a particular composition.

6.1 A weighted block-moment estimate

In a composition belonging to block k , let $D = \sum_i d_{m_i+1}$ and put $A_k = k \log(k+1)$. Since $\sum_i (m_i + 1) \leq 2k$ and every $m_i + 1 \leq k$,

$$\sum_i L_{m_i+1} \leq 2k \log(k+1) = 2A_k.$$

For nonnegative numbers and $r \geq 1$, $\sum x_i^r \leq (\sum x_i)^r$. Consequently

$$\sum_i \lambda_{m_i+1} \leq K^r (2A_k + D)^r.$$

The subtraction in (5.4) has magnitude at most $2k\lambda_1$. Increasing a constant also covers block $k = 1$, and gives

$$|b_{k,\mathbf{m}}| \leq C_s (A_k + D)^r. \quad (6.1)$$

For every integer $m \geq 1$,

$$\begin{aligned} e^{-D} \frac{|b|^m}{m!} &\leq \frac{C_s^m}{m!} e^{-D} (A_k + D)^{rm} \\ &\leq \frac{E_s^m}{m!} \left(A_k^{rm} + \left(\frac{rm}{e} \right)^{rm} \right). \end{aligned} \quad (6.2)$$

Here we used $(x+y)^{rm} \leq 2^{rm}(x^{rm} + y^{rm})$ and $\sup_{D \geq 0} e^{-D} D^{rm} = (rm/e)^{rm}$. The case $m = 0$ is bounded by one. Thus amplitude damping converts a possibly unbounded frequency into a factorially controlled moment. Bounding frequencies alone would miss this mechanism.

Lemma 6.1 (Factorial budget). *For each $s > 0$ there is $B_s \geq 1$ such that, for $k \geq 1$, $m \geq 0$, and $n = k + m$,*

$$\frac{[k \log(k+1)]^{(s+1)m}}{m!} + \frac{m^{(s+1)m}}{m!} \leq B_s^n (n!)^s. \quad (6.3)$$

In the case $m = 0$, powers with exponent zero are interpreted as one.

Proof. Write $r = s + 1$. The second summand follows at once from $m! \geq (m/e)^m$ and $n! \geq (n/e)^n$, since $m^{rm}/m! \leq e^m m^{sm} \leq e^{(s+1)n} (n!)^s$.

For the first summand assume $m \geq 1$, and use $\log h! \geq h \log h - h$. Its logarithm after division by $(n!)^s$ is at most

$$rm \log(A_k/n) - sk \log n - m \log(m/n) + m + sn.$$

The last three terms other than $-sk \log n$ are $O_s(n)$, because $-x \log x$ is bounded on $[0, 1]$. Put $x = (k/n) \log(n+1)$. Then $A_k/n \leq x$, and $k \log n \geq nx/2$ for $n \geq 2$. The remaining expression is at most

$$n \left(r \log^+ x - \frac{s}{2} x + O_s(1) \right).$$

The function $r \log^+ x - sx/2$ is bounded above on $[0, \infty)$ because $s > 0$. Exponentiation proves the assertion. The finitely many omitted cases are absorbed into B_s . $\square$

6.2 Coefficient sufficiency

Extracting the ordinary coefficient of u^n from (5.1) uses $k \leq n$ and $m = n - k$. Apply (6.2), then Lemma 6.1, and finally the mass bound 8^k . The sum over the n possible blocks is at most $CH^n(n!)^s$. It follows that $\widehat{Q} \in \mathcal{G}_s$. Lemma 3.1 gives $\widehat{U} \in \mathcal{G}_s$. This proves sufficiency for both formal series without using any asymptotic formula for individual coefficients.

6.3 The same budget controls an actual analytic remainder

For every complex z and integer $m \geq 1$, the integral form of the exponential Taylor remainder gives

$$\left| e^z - \sum_{h=0}^{m-1} \frac{z^h}{h!} \right| \leq \frac{|z|^m}{m!} e^{\max(0, \operatorname{Re} z)}. \quad (6.4)$$

For $\operatorname{Re} u \leq 0$ and a real block frequency b , the exponential factor on the right is at most one if $b \geq 0$. If $b < 0$, it is at most $e^{2k\lambda_1\tau_0}$. This latter factor is merely exponential in k .

In every block $k < N$, truncate e^{bu} before degree $N - k$. The resulting finite sum is exactly $\sum_{n < N} v_n u^n$. By (6.4) the error in block k is bounded by $|u|^N$ times the same moment in (6.2), with $m = N - k$, and an extra fixed exponential in k . The factorial budget and mass bound therefore give a total at most $CH^N(N!)^s|u|^N$.

For the blocks $k \geq N$, choose r_0 as in Section 5. Their geometric tail is bounded by $CH_0^N|u|^N$. Adding the two bounds proves (2.5), including its boundary values. The case $N = 1$ follows directly from the geometric series.

Conversely, if (2.5) holds, its order- $N + 1$ version shows along the negative ray that

$$\frac{Q(u) - \sum_{n < N} v_n u^n}{u^N} \rightarrow v_N.$$

Its order- N version then gives $|v_N| \leq CA^N(N!)^s$. Thus a uniform expansion implies $\hat{Q} \in \mathcal{G}_s$, and the necessity argument of Section 3 applies after reversion. This completes the Gevrey part of Theorem 2.2.

Remark 6.2 (Why the analytic criterion is different). The proof of Lemma 6.1 uses $s > 0$ in an essential way. Setting $s = 0$ removes the term that offsets $k \log(k + 1)$. Accordingly, the convergence budget is $j + d_j$, not the result of formally putting $s = 0$ in the positive-Gevrey budget. There are divergent series belonging to every $\mathcal{G}_s$, $s > 0$.

7 Bessel kernels and fine negative-ray summation

7.1 One exact Borel kernel

For an integer $k \geq 1$ and a real number b , define the entire function

$$K_{k,b}(\zeta) = \sum_{m \geq 0} \frac{b^m \zeta^{k+m}}{m!(k+m)!} = \mathcal{B}_0(u^k e^{bu})(\zeta). \quad (7.1)$$

These are Bessel kernels in elementary power-series normalization. Their relation to the standard I_k and the Poisson integral for J_k can be read from the formulas in [6]. The following direct version supplies exactly the estimates we need.

Let ν_k be the probability measure on $[-1, 1]$ with density proportional to $(1 - t^2)^{k-1/2}$. Its odd moments vanish and the beta integral gives

$$\int t^{2m} d\nu_k(t) = \frac{(2m)!k!}{4^m m!(k+m)!}.$$

Expanding the exponential, and integrating on this compact interval, therefore proves

$$K_{k,b}(\zeta) = \frac{\zeta^k}{k!} \int_{-1}^1 \exp(2\sqrt{b\zeta} t) d\nu_k(t). \quad (7.2)$$

The expression is independent of the square-root choice. Put

$$h(\zeta) = |\operatorname{Re} \sqrt{\zeta}|, \quad \tilde{h}(\zeta) = |\operatorname{Im} \sqrt{\zeta}|, \quad h(\zeta)^2 = \frac{|\zeta| + \operatorname{Re} \zeta}{2}.$$

Equation (7.2) implies

$$|K_{k,b}(\zeta)| \leq \frac{|\zeta|^k}{k!} \begin{cases} \exp(2h(\zeta)\sqrt{b}), & b \geq 0, \\ \exp(2\tilde{h}(\zeta)\sqrt{|b|}), & b < 0. \end{cases} \quad (7.3)$$

In particular, for $t \geq 0$ and $b \geq 0$, oscillation gives $|K_{k,b}(-t)| \leq t^k/k!$.

For one fixed b and $r > 0$, termwise Laplace integration is justified absolutely and yields

$$\frac{1}{r} \int_0^\infty e^{-t/r} K_{k,b}(-t) dt = (-r)^k e^{-br}. \quad (7.4)$$

With an absolute value inside the integral, its value is at most r^k for $b \geq 0$, by the oscillatory bound, and is $r^k e^{|b|r}$ for $b < 0$, by the positive Taylor series. This stronger distinction will justify an infinite exchange that termwise absolute Taylor expansion would not.

7.2 The all-slope ray transform is not automatically a Borel germ

Write the internal block sums as in (5.1) and set

$$\mathfrak{B}_Q(-t) = \sum_{k \geq 1} \sum_{\mathbf{m}} \omega_{k,\ell} e^{-D\mathbf{m}} K_{k,b_{k,\mathbf{m}}}(-t). \quad (7.5)$$

The absolute contribution of block k is bounded by

$$8^k \frac{t^k}{k!} \exp(2\sqrt{2k\lambda_1}t).$$

Young's inequality bounds the last exponent by $\eta k + C_\eta t$. Summing the resulting exponential series proves continuity on the ray and a bound Ce^{Ht} . Moreover, (7.4) bounds the integrated absolute contribution of block k by $8^k r^k e^{2k\lambda_1 r}$. Hence Fubini's theorem gives

$$Q(-r) = \frac{1}{r} \int_0^\infty e^{-t/r} \mathfrak{B}_Q(-t) dt \quad (7.6)$$

for sufficiently small r . This construction works for every slope sequence. Without further growth control, however, $\mathfrak{B}_Q$ need not extend to a holomorphic germ at the origin. Equation (7.6) alone is not a Borel-summability theorem.

7.3 Amplitude damping gives a complex neighborhood

Now assume the joint criterion with $s = 1$. Equation (6.1) gives, for positive frequencies,

$$\sqrt{b} \leq C(A_k + D), \quad A_k = k \log(k+1).$$

Combining this with (7.3) retains the amplitude cost:

$$e^{-D} |K_{k,b}(\zeta)| \leq \frac{|\zeta|^k}{k!} e^{2Ch(\zeta)A_k} e^{-(1-2Ch(\zeta))D}, \quad b \geq 0. \quad (7.7)$$

For negative frequencies use $|b| \leq 2k\lambda_1$ instead. On any compact set with $2Ch < 1$, the absolute mass of block k is consequently at most

$$\frac{8^k |\zeta|^k}{k!} \exp\left(2Ch(\zeta)k \log(k+1) + 2\sqrt{2\lambda_1} \tilde{h}(\zeta) \sqrt{k}\right). \quad (7.8)$$

Its logarithm is $-(1-\theta)k \log k + O(k) + O(\sqrt{k})$ for a fixed $\theta < 1$ on that compact set. The series

$$B_Q(\zeta) = \sum_{k \geq 1} \sum_{\mathbf{m}} \omega_{k,\ell} e^{-D\mathbf{m}} K_{k,b_{k,\mathbf{m}}}(\zeta) \quad (7.9)$$

therefore converges normally on $\{2Ch < 1\}$. Its Taylor coefficient of order n uses only blocks $k \leq n$ and equals $v_n/n!$. Thus it is indeed the analytic continuation of the Borel germ of $\hat{Q}$.

7.4 Growth on a fixed-width tube

Choose $\delta > 0$ so small that the closure of $\Omega_{2\delta}^-$ is contained in $\{2Ch < 1\}$. For $\zeta = -t + iy$, $t \geq 1$, $|y| \leq \delta$,

$$h(\zeta) \leq \frac{\delta}{2\sqrt{t}}, \quad \tilde{h}(\zeta) \leq \sqrt{t + \delta}, \quad |\zeta| \leq t + \delta.$$

The logarithm of (7.8) is bounded by

$$k \log(8(t + \delta)) - \log k! + \frac{C_1}{\sqrt{t}} k \log(k + 1) + C_2 \sqrt{kt}. \quad (7.10)$$

If $k \leq t^2$, the third term is $o(k)$ as $t \rightarrow \infty$ and $C_2 \sqrt{kt} \leq k + C_3 t$. Summing these terms gives a bound $Ce^{H_1 t}$. If $k > t^2$, then $\log(t + \delta) \leq \frac{1}{2} \log k + o(1)$. For sufficiently large t , $C_1/\sqrt{t} < 1/8$ and $C_2 \sqrt{t/k}$ is small. The factorial estimate $\log k! \geq k \log k - k$ then bounds (7.10) by $-\frac{1}{4} k \log k$. This tail is uniformly bounded. The remaining bounded part of the tube is compactly contained in the normal-convergence domain. It follows that

$$|B_Q(\zeta)| \leq Ce^{H|\zeta|}, \quad \zeta \in \Omega_\delta^-. \quad (7.11)$$

This proves fine summability of $\hat{Q}$ and identifies its sum with the literal Q by (7.6).

7.5 Passing from the inverse to the forward series

Fine summability is closed under compositional inversion for a series with linear coefficient one. Because fine and angular summability are different hypotheses, Appendix A proves the precise fixed-tube version used here. Its proof is a convolution argument in the Borel plane, following the classical framework in [5]. It shows simultaneously that summation commutes with inversion.

Applying that result to $\hat{Q}$ proves fine summability of $\hat{U}$ and the identity between the two summed inverse functions. The literal equation follows from (5.7), on the small common domain where the dependent variable has negative real part. Conversely, fine summability implies a holomorphic Borel germ and therefore Gevrey-1 coefficients. Theorem 2.2 then gives the joint budget. This completes Theorem 2.4.

8 Exact forward type in the strong-damping regime

The next argument is independent of the Borel construction. It uses positivity of the forward Lagrange weights, and it controls their exponential scale sharply enough to compute an exact type.

8.1 A universal upper-type lemma

Lemma 8.1 (A power bound on slopes in cost coordinates). *Fix $r > 1$, put $s = r - 1$, and suppose, for some $a > 0$ and $C \geq 0$, that*

$$\lambda_j \leq ad_j^r + C \quad (j \geq 1). \quad (8.1)$$

Then $T_s(\hat{U}) \leq ar^r$. The same upper bound applies to the sum of any subcollection of the nonnegative Lagrange terms satisfying the analogous bound $\sum_i \lambda_{j_i} \leq aD^r + CK$.

Proof. For a length- k composition of n let $D = \sum_i d_{j_i}$ and $\Lambda = \sum_i \lambda_{j_i}$. Since $\sum_i d_{j_i}^r \leq D^r$, we have $\Lambda \leq aD^r + Ck$. Increasing C to a positive constant permits the harmless bound

$$\frac{e^{-D} \Lambda^{k-1}}{k!} \leq \frac{1}{k!} e^{-D} (aD^r + Cn)^k.$$

There are $\binom{n-1}{k-1}$ ordered compositions. Define

$$M_{n,k} = \frac{1}{k!} \sup_{D \geq 0} e^{-D} (aD^r + Cn)^k.$$

A first, deliberately coarse estimate follows from $(x+y)^k \leq 2^k(x^k + y^k)$:

$$M_{n,k} \leq \frac{2^k}{k!} \left(a^k (rk)^{rk} e^{-rk} + (Cn)^k \right). \quad (8.2)$$

The logarithm of the first summand is at most $sk \log k + O(n)$, uniformly for $1 \leq k \leq n$; the second is at most $e^{O(n)}$, since $(Cn)^k/k! \leq e^{Cn}$. After multiplying by at most 2^n compositions and summing over $k \leq (1-\delta)n$, division by $(n!)^s$ therefore leaves

$$\exp(-s\delta n \log n + O(n)) \quad (8.3)$$

for every fixed $\delta \in (0, 1)$.

For $k \geq (1-\delta)n$, with $\delta < 1/2$, set $D = kx$. Then

$$\begin{aligned} \sup_D e^{-D} (aD^r + Cn)^k &= k^{rk} \left[\sup_{x \geq 0} e^{-x} \left(ax^r + \frac{Cn}{k^r} \right) \right]^k \\ &\leq k^{rk} (a(r/e)^r + o(1))^k, \end{aligned}$$

where $o(1)$ is uniform in this range of k , since $r > 1$. Stirling's estimate now gives

$$M_{n,k} \leq e^{o(n)} (ar^r + o(1))^k (k!)^s. \quad (8.4)$$

The sum of the numbers of compositions in this range is at most $e^{n\mathfrak{h}(\delta)+o(n)}$, where $\mathfrak{h}(\delta) = -\delta \log \delta - (1-\delta) \log(1-\delta)$. Using $k! \leq n!$, the n th root of the normalized contribution is at most

$$e^{\mathfrak{h}(\delta)} \max_{1-\delta \leq t \leq 1} (ar^r)^t + o(1).$$

First let $n \rightarrow \infty$ and then $\delta \downarrow 0$. Together with (8.3), this proves $T_s(\widehat{U}) \leq ar^r$. All the estimates are upper bounds on positive summands and therefore remain valid for a restricted subcollection. $\square$

The loss in (8.3) is stronger than exponential. The sharp type comes from compositions whose length is $n - o(n)$; their entropy has vanishing rate. This is why replacing every composition count by 2^n throughout would destroy the exact constant.

8.2 A matching one-action lower bound

Suppose (2.8) holds. We first record the implication

$$j = o(d_j), \quad j \log d_j = o(d_j). \quad (8.5)$$

Indeed write $d_j = M_j j \log(j+1)$ with $M_j \rightarrow \infty$. Since $\log x/x$ is decreasing for large x , or directly from this expression,

$$\frac{j \log d_j}{d_j} = \frac{\log M_j + \log j + \log \log(j+1)}{M_j \log(j+1)} \rightarrow 0.$$

Fix $r = s+1$ and put $A = \limsup_j \lambda_j/d_j^r$. For every $a > A$ when $A < \infty$, the finite exceptions can be included in a constant C so that (8.1) holds. Lemma 8.1 proves $T_s(\widehat{U}) \leq ar^r$; taking $a \downarrow A$ also covers $A = 0$.

For the lower bound take any $B > 0$ smaller than A , or any $B > 0$ if $A = \infty$. There are arbitrarily large j with $\lambda_j \geq Bd_j^r$. Put $n_j = \lfloor d_j/r \rfloor$ and $m_j = n_j - j$. Equations (8.5) ensure $m_j \geq 0$ eventually, $m_j/n_j \rightarrow 1$, and $j \log n_j/n_j \rightarrow 0$. The one-action term gives

$$\frac{u_{n_j}}{(n_j!)^{r-1}} \geq \frac{e^{-d_j}(Bd_j^r)^{m_j}}{m_j!(n_j!)^{r-1}}.$$

Stirling's estimate, $d_j/n_j \rightarrow r$, and $j \log n_j/n_j \rightarrow 0$ imply

$$\frac{1}{n_j} \log \left(\frac{e^{-d_j}(Bd_j^r)^{m_j}}{m_j!(n_j!)^{r-1}} \right) = \log B + r \log r + o(1). \quad (8.6)$$

Thus $T_s(\widehat{U}) \geq Br^r$. Letting $B \uparrow A$ proves Theorem 2.5 for finite positive A , and arbitrary B proves it when $A = \infty$. When $A = 0$ the upper bound already gives equality. No density assumption on the sequence of costs was used: a subsequence is enough for a limsup.

8.3 What the type identity does and does not say

The identity allows isolated large slopes and arbitrarily irregular costs. In particular, finite changes to amplitudes or slopes do not change this exact type, although they do change the coefficients and can change subexponential factors substantially. A type theorem gives only $\limsup(u_n/(n!)^s)^{1/n}$; it need not give a limit and never by itself gives a multiplicative equivalent for u_n . Signed reversion coefficients require a different argument, so no inverse-type identity is inferred here.

9 Dense power costs and a dominant-action limit

Assume (2.10). Let $r = p/\beta$, $s = r - 1$, and $A = a/b^r$. Then $d_j/(j \log(j+1)) \rightarrow \infty$ and $\lambda_j/d_j^r \rightarrow A$. Theorem 2.5 already gives an upper coefficient-root limit Ar^r .

9.1 Obtaining the lower bound at every degree

For each large n , choose an integer j_n nearest to $(rn/b)^{1/\beta}$. Since $\beta > 1$,

$$j_n = o(n), \quad j_n \log n = o(n), \quad d_{j_n}/n \rightarrow r, \quad \lambda_{j_n}/d_{j_n}^r \rightarrow A.$$

The one-action lower bound with $m = n - j_n$ now applies at every degree n . The calculation in (8.6) gives

$$\liminf_n \left(\frac{u_n}{(n!)^s} \right)^{1/n} \geq Ar^r.$$

This proves (2.11). Equivalently,

$$\log u_n = s \log(n!) + n \log(Ar^r) + o(n). \quad (9.1)$$

It is not the assertion $u_n \sim (Ar^r)^n (n!)^s$.

9.2 The probability space and the length concentration

For an ordered composition $\mathbf{j} = (j_1, \dots, j_k)$ of n , define

$$\mathbb{P}_n(\mathbf{j}) = \frac{1}{u_n k!} e^{-\sum_i d_{j_i}} \left(\sum_i \lambda_{j_i} \right)^{k-1}. \quad (9.2)$$

Formula (3.1) says these probabilities sum to one. A term with zero weight has zero probability. The variable K is the number of parts, and D is their total amplitude cost.

For every fixed $\delta > 0$, (8.3) and the all-degree lower bound (9.1) imply

$$\mathbb{P}_n\{K \leq (1 - \delta)n\} \leq \exp(-s\delta n \log n + O(n) + o(n)) \longrightarrow 0. \quad (9.3)$$

Hence $K/n \rightarrow 1$. This does not say there is only one nonunit part; it says the aggregate loss in the number of parts is sublinear.

9.3 Total cost is concentrated at the unique variational maximum

The scalar function

$$x \longmapsto e^{-x} x^r \quad (9.4)$$

has its unique maximum $(r/e)^r$ at $x = r$. For every fixed $\varepsilon > 0$, its supremum on $|x - r| \geq \varepsilon$ is strictly smaller. This remains a strict gap after adding a quantity tending uniformly to zero, as in the proof of (8.4).

Fix a tolerance $\varepsilon > 0$ for D/n . Restrict to $K/n \geq 1 - \delta$, with δ sufficiently small. If $|D/n - r| \geq \varepsilon$, then $|D/K - r| \geq \varepsilon/2$ after possibly shrinking δ ; this follows separately for the upper and lower inequalities. In the sharp-type proof, the supremum over $x = D/K$ is therefore restricted away from r . It loses a fixed exponential factor. Use a slope bound with coefficient $a_+ > A$ as close to A as needed, and then take δ so small that the composition entropy $\mathfrak{h}(\delta)$ is smaller than that loss. The contribution has normalized coefficient-root upper limit strictly below Ar^r . Equation (9.1) makes its probability tend to zero. The excluded range $K/n < 1 - \delta$ is negligible by (9.3). Thus $D/n \rightarrow r$ in probability.

For clarity, the two choices are compatible: the continuous function in (9.4) has a positive gap for the fixed tolerance; choose $a_+ - A$ first small compared with that gap and then choose δ small compared with both the gap and the tolerance. The tail $x \rightarrow \infty$ causes no difficulty because (9.4) tends to zero.

9.4 A single part carries asymptotically all the cost

Let $M = \max_i d_{j_i}$. On the event $M \leq (1 - \varepsilon)D$,

$$\sum_i d_{j_i}^r \leq M^{r-1} D \leq (1 - \varepsilon)^{r-1} D^r. \quad (9.5)$$

Write $\gamma = (1 - \varepsilon)^{r-1} < 1$. Since $\lambda_j \leq a_+ d_j^r + C$ for every $a_+ > A$, the slopes on this event satisfy $\Lambda \leq a_+ \gamma D^r + CK$. The restricted-sum version of Lemma 8.1 gives an upper type $a_+ \gamma r^r$. Choose a_+ close enough to A that $a_+ \gamma < A$. The all-degree root limit then shows that the probability of the event tends to zero exponentially. As $D/n \rightarrow r > 0$, the event $D = 0$ is negligible. Therefore $M/D \rightarrow 1$ in probability.

It follows that $M/n \rightarrow r$. The asymptotic $d_j \sim b j^\beta$ implies $M/(b J_{\max}^\beta) \rightarrow 1$ whenever $J_{\max} \rightarrow \infty$: finite initial indices contribute a bounded cost, and the asymptotic bounds apply uniformly to all larger indices. Since $M/n \rightarrow r$, that divergence holds in probability. Consequently

$$J_{\max}/n^{1/\beta} \rightarrow (r/b)^{1/\beta}.$$

This completes every assertion of Theorem 2.6.

Remark 9.1 (Meaning of condensation here). The proved concentration concerns a probability distribution on the positive Lagrange contributions, not a random solution of the feedback equation. One action carries almost all amplitude cost and hence almost all of the power-law slope budget. The theorem gives neither its local limit law nor the distribution of the remaining small actions. Those would be needed for a sharp multiplicative coefficient asymptotic.

10 A sharper Borel region under strong damping

Proposition 10.1 (Damping-controlled inverse continuation). *Assume (2.8) and let $A = \limsup_j \lambda_j/d_j^2 < \infty$. If $A > 0$, the explicit inverse Borel series (7.9) converges normally on*

$$\mathcal{P}_A = \left\{ \zeta : |\zeta| + \operatorname{Re} \zeta < \frac{1}{2A} \right\}. \quad (10.1)$$

If $A = 0$, it defines an entire function. In both cases it has at most exponential growth on a sufficiently thin fixed-width tube around the negative ray. When $0 < A < \infty$, the forward Borel transform has radius of convergence exactly $1/(4A)$ and has a singularity at the positive point $1/(4A)$. Also $T_1(\widehat{Q}) \leq 4A$; equality and a singularity of the inverse at the same point are not asserted.

Proof. For every $a > A$ there is a constant C with $\lambda_j \leq ad_j^2 + C$. Hence every positive inverse-block frequency satisfies $b \leq aD^2 + Ck$ and $\sqrt{b} \leq \sqrt{a}D + \sqrt{Ck}$. Equation (7.3) now gives

$$e^{-D} |K_{k,b}(\zeta)| \leq \frac{|\zeta|^k}{k!} e^{-(1-2\sqrt{a}h(\zeta))D} e^{2\sqrt{C}h(\zeta)\sqrt{k}}.$$

On compact sets with $2\sqrt{a}h < 1$, the first exponential is at most one. Negative frequencies contribute at most $\exp(2\sqrt{2\lambda_1}\tilde{h}\sqrt{k})$ instead. The mass bound and the factorial in the denominator prove normal convergence. Taking the union over $a > A$ gives (10.1); for $A = 0$, every compact subset of the plane is permitted by taking a small enough.

On a fixed tube, h is bounded and $\tilde{h} = O(\sqrt{1+|\zeta|})$. Young's inequality absorbs the remaining square-root terms into $C_1k + C_2(1+|\zeta|)$, and the sum over k is exponentially bounded. This argument is simpler than (7.10), because the strong cost estimate has removed $k \log k$ from the positive-frequency bound.

Theorem 2.5 at $s = 1$ gives $T_1(\widehat{U}) = 4A$, hence the stated forward Borel radius. A positive-coefficient power series of finite radius R cannot be analytic at R : if it were, take a real $x < R$ close enough to R that its Taylor expansion about x converges at some $y > R$. Its Taylor coefficients at x are nonnegative. Interchanging the resulting nonnegative double sum would show that the original power series converges at y , a contradiction. This classical positive-coefficient argument proves the forward singularity. Finally, (10.1) contains every disc of radius less than $1/(4A)$, so the inverse Borel radius is at least $1/(4A)$. $\square$

The positive intercept of the guaranteed inverse parabola is therefore exactly the forward Borel radius. This is a useful alignment of two independent calculations, but it does not prove that the boundary of (10.1) is a natural boundary, or even that every point on it is singular for the inverse. Normal convergence regions and maximal continuation domains must not be identified.

11 Phase diagrams and explicit examples

11.1 Power-law slopes and stretched-exponential amplitudes

Suppose $d_j \sim bj^\beta$ and $\lambda_j \sim aj^p$, with $a, b, \beta, p > 0$, and keep the normalization at $j = 1$. Put $B = \max(1, \beta)$. The sharp classification reads

Property	Necessary and sufficient condition
Ordinary convergence	$p \leq B$
Gevrey- s , for fixed $s > 0$	$p \leq (s + 1)B$
Fine summability in direction π	$p \leq 2B$
Least Gevrey order when $p > B$	$s_* = p/B - 1$, attained

At $\beta \leq 1$, the logarithm in $j \log(j + 1)$ makes every equality case in the Gevrey row permissible. For $\beta > 1$ the damping cost dominates that logarithm. In the divergent range $p > \beta > 1$,

$$T_{p/\beta-1}(\widehat{U}) = a \left(\frac{p}{b\beta} \right)^{p/\beta}, \quad J_{\max} \sim_{\mathbb{P}} \left(\frac{p}{b\beta} n \right)^{1/\beta}. \quad (11.1)$$

The notation $\sim_{\mathbb{P}}$ denotes a ratio tending to one in probability under (9.2).

Example 11.1 (Quadratic costs and cubic slopes). Let $c_j = 2^{-j^2}$ and $\lambda_j = j^3$ for $j \geq 2$, with $c_1 = 1$ and $\lambda_1 = 1$. Here $\beta = 2$, $b = \log 2$, $p = 3$. The formal solution is divergent, has least Gevrey order $1/2$, and satisfies

$$\lim_n \left(\frac{u_n}{\sqrt{n!}} \right)^{1/n} = \left(\frac{3}{2 \log 2} \right)^{3/2}.$$

Both series are finely summable in the negative direction, since they are in particular Gevrey-1. Their ordinary Borel transforms are entire in this case: for the forward series this follows from its order-1/2 bound, and for the inverse from Proposition 10.1 with $A = 0$. Entire does not imply exponential growth in every direction.

Example 11.2 (A nonzero critical Borel scale). For $j \geq 2$, take $c_j = \exp(-bj^\beta)$ and $\lambda_j = aj^{2\beta}$, where $\beta > 1$ and $a, b > 0$. The least Gevrey order is one, with exact forward type $4a/b^2$. The forward Borel transform has its first positive singularity at $\zeta = b^2/(4a)$. The inverse has the guaranteed continuation region

$$|\zeta| + \operatorname{Re} \zeta < \frac{b^2}{2a}.$$

It is finely summable on the negative ray. Nothing in this argument classifies the singularity at the positive point or its Stokes data.

11.2 Logarithmic boundary factors

The budgets also settle endpoints not visible in a pure power diagram. Suppose, for $j \rightarrow \infty$,

$$d_j \sim bj^\beta (\log j)^\gamma, \quad \lambda_j \sim aj^p (\log j)^\eta, \quad a, b, \beta, p > 0, \quad \gamma, \eta \in \mathbb{R}.$$

Order pairs lexicographically. Let

$$(B_0, G_0) = \max_{\text{lex}} \{(1, 0), (\beta, \gamma)\}, \quad (B_1, G_1) = \max_{\text{lex}} \{(1, 1), (\beta, \gamma)\}.$$

Comparison of positive monomials gives the complete tests

$$\widehat{U} \text{ converges} \iff (p, \eta) \leq_{\text{lex}} (B_0, G_0), \quad (11.2)$$

$$\widehat{U}, \widehat{Q} \in \mathcal{G}_s \iff \left(\frac{p}{s+1}, \frac{\eta}{s+1} \right) \leq_{\text{lex}} (B_1, G_1). \quad (11.3)$$

The second test at $s = 1$ is also the fine-summability test. For example, when $d_j \sim bj^\beta$ with $\beta > 1$ and $p > \beta$, the infimal positive order is $p/\beta - 1$. At that order a factor $(\log j)^\eta$ in the slopes is permitted exactly when $\eta \leq 0$. Thus an infimal Gevrey order need not be attained in the general logarithmic setting. The main criterion records that distinction.

11.3 Superpolynomial slopes can still be summable

Example 11.3 (Exponential slopes, exact Gevrey-1). For $j \geq 2$ set

$$c_j = \exp(-e^{j/2}), \quad \lambda_j = e^j,$$

and set $c_1 = 1$, $\lambda_1 = 1$. Then $d_j = e^{j/2}$ and $\lambda_j = d_j^2$. The convergence criterion fails, but the least Gevrey order is exactly one and $T_1(\widehat{U}) = 4$. The forward Borel radius is $1/4$; the inverse has the guaranteed region $|\zeta| + \operatorname{Re} \zeta < 1/2$; both series are finely summable in direction π . The geometric spacing of these costs does not meet the power-cost hypothesis of Theorem 2.6, so no all-degree root limit is claimed by that theorem.

Example 11.4 (Damping sufficient for ordinary convergence). Let $\lambda_j \geq 0$ be any sequence, however fast it grows, and take $c_j = \exp(-\lambda_j)$ for $j \geq 2$, with $c_1 = 1$. Then $d_j = \lambda_j$ for $j \geq 2$, so $\lambda_j/(j + d_j) \leq 1$. The formal solution is genuinely convergent. The amplitude cost can therefore restore ordinary analyticity, not just a weaker factorial regularity. Finitely many initial slopes do not affect this conclusion.

Example 11.5 (A divergent series in every positive Gevrey class). Let $c_j = 1$ and $\lambda_j = j \log(j+1)$. Then $\lambda_j/j \rightarrow \infty$, so the solution diverges. For every $s > 0$, $\lambda_j^{1/(s+1)}/(j \log(j+1))$ is bounded. Thus the solution and its inverse belong to every positive Gevrey class and are finely summable in direction π . This is an illustration of the distinction between the two budgets, not a new unit-amplitude theorem.

12 Why the hypotheses and the direction matter

12.1 Exponential amplitude control is a substantive assumption

The joint fine-summability equivalence is not a theorem for arbitrary positive amplitudes. Take $\lambda_j = 0$ for all j and

$$c_j = j!(2 + (-1)^j), \quad j \geq 1.$$

Then $c_1 = 1$ and the equation is simply $\widehat{U}(q) = \sum c_j q^j$. The series is Gevrey-1, but its integrated Borel transform is

$$\mathcal{B}_0 \widehat{U}(\zeta) = \frac{2\zeta}{1-\zeta} - \frac{\zeta}{1+\zeta}. \quad (12.1)$$

The pole at $\zeta = -1$ prevents fine summability in the negative direction. The amplitudes are not exponentially bounded, so this does not contradict Theorem 2.4. It shows exactly where the geometric block mass used in the proof can fail.

12.2 Positivity is not just a convenient estimate

The one-action lower bound, the exact-type lower bound, and the probability interpretation use nonnegative Lagrange terms. Signed or complex amplitudes can cancel those terms. A theorem for such models would need a noncancellation condition or different invariants. Similarly, negative or complex slopes destroy the common damping half-plane used to sum the literal kernel and to control Bessel frequencies. The present necessity-and-sufficiency statements should not be extended to those settings by taking absolute values in their conclusions.

12.3 Why positive-ray summation fails for divergent positive series

The asymmetry between positive and negative directions is intrinsic. Suppose $\widehat{U}$ has nonnegative coefficients, is divergent, and is Gevrey-1. If its Borel transform has finite radius, the positive-coefficient argument in Proposition 10.1 puts a singularity on the positive ray. If its Borel

transform is entire and had an exponential bound $\mathcal{B}_0\widehat{U}(t) \leq Ce^{Ht}$ for $t \geq 0$, positivity would give

$$\frac{u_n}{n!} \leq \frac{Ce^{Ht}}{t^n}.$$

Taking $t = n/H$ (and increasing H to be positive if necessary) and using Stirling’s estimate would yield $u_n \leq C'\sqrt{n}H^n$, forcing ordinary convergence. This is a contradiction. Thus a divergent positive solution cannot be finely Borel summable in the positive direction, even in cases where its Borel transform is entire.

12.4 Remaining distinctions

The literal inverse is analytic in a small open left half-disc and has continuous boundary values supplied by its blocks. The uniform Gevrey– s remainder theorem is proved for that inverse. For $s = 1$, fine summation gives a canonical value in its summation class and an inverse forward sum. These statements do not imply a common open arc of summation directions, resurgent continuation along arbitrary paths, a sharp least-truncation error, or a Stokes formula.

For $s \neq 1$ the factorial remainder theorem does not, by itself, prove summability with a moment kernel of order s . In particular, the all-slope continuous ray transform in (7.5) is not to be relabeled as a holomorphic Borel transform when its germ fails to exist. The separation of these assertions is part of the theorem.

13 Exact checks, numerical diagnostics, and reproducibility

13.1 Independent finite identities

The package’s primary verification script, `code/verify.py`, uses exact rational arithmetic from Python’s standard library. Its recorded run completed **1,353 assertions** at coefficient order 12 across twelve models. The models combine several integer-power slope sequences with unit, geometric, quadratic-cost, and irregular positive rational amplitudes. The checks compare two independent forward constructions: the exponential coefficient recurrence and the ordered Lagrange formula. The inverse is constructed from the signed exponential blocks, after which both compositions $\widehat{U} \circ \widehat{Q}$ and $\widehat{Q} \circ \widehat{U}$ are checked coefficientwise.

For reference, the recurrence used independently of Lagrange inversion is as follows. Write

$$E_j(q) = \exp(\lambda_j \widehat{U}(q)) = \sum_{k \geq 0} e_{j,k} q^k.$$

Differentiation and coefficient comparison give

$$e_{j,0} = 1, \quad e_{j,k} = \frac{\lambda_j}{k} \sum_{m=1}^k m u_m e_{j,k-m}, \quad u_n = \sum_{j=1}^n c_j e_{j,n-j}. \quad (13.1)$$

Every $e_{j,n-j}$ on the right uses only previously computed u_m . No nonlinear numerical root finder is needed. The other exact checks cover the one-action lower bound and the block mass bound through block 100. The certificate below uses explicit rational exponential enclosures together with checks on its domain and final width. Exact finite checks test identities and implementation consistency; they are not proofs of the asymptotic statements.

13.2 A rational certificate for a literal inverse value

For the cubic-slope, quadratic-cost model

$$c_1 = 1, \quad c_j = 2^{-j^2} \ (j \geq 2), \quad \lambda_j = j^3,$$

the script encloses the first ten inverse blocks at $u = -1/100$ using rational exponential Taylor bounds and adds the geometric tail (5.8). It produces the following outward-rounded decimal enclosure:

$$-0.010106452308614 < Q(-0.01) < -0.010106452306282. \quad (13.2)$$

The file `data/exact_checks.json` records the exact rational endpoints. Their interval has width less than 2.331×10^{-12} ; the displayed decimals are rounded outward. This certifies the literal implicit branch, rather than agreement between two floating-point implementations. The tail bound ignores cancellation and is not optimized.

An independent interval evaluation of the literal kernel, using its first 15 actions and a geometric tail, confirms opposite residual signs at these rational endpoints. On $|q| \leq 1/50$, its derivative is bounded below by

$$1 - \frac{1}{100} - \left((1 - 1/50)^{-2} - 1 \right) > 0.$$

Thus the same interval contains a unique literal root by a second route. The endpoint sign enclosures and derivative bound are recorded alongside the block certificate.

13.3 Coefficient-root diagnostics

The separate script `code/diagnostics.py` evaluates (13.1) in logarithmic floating-point arithmetic, using NumPy and SciPy. It was first compared with exact rational coefficients through degree 20; the largest discrepancy between logarithms was 4.27×10^{-14} after upward rounding. The subsequent calculations through degree 512 are diagnostics, not interval certificates.

The table uses $c_j = 2^{-j^2}$ for $j \geq 2$, $c_1 = 1$, $\lambda_j = j^p$, and

$$R_n = \frac{(u_n/(n!)^{p/2-1})^{1/n}}{(p/(2 \log 2))^{p/2}}.$$

Theorem 2.6 gives $R_n \rightarrow 1$.

n	$p = 3$	$p = 4$	$p = 5$
64	0.592248	0.305540	0.149857
128	0.658251	0.384842	0.213325
256	0.719314	0.468507	0.290177
512	0.773542	0.551826	0.376472

Convergence is visibly slow at these degrees. The discrepancy is not an error bar for the theorem and is not evidence for a multiplicative coefficient equivalent. The recorded diagnostic file additionally contains large-degree one-action lower bounds, which are inexpensive to compute but are not the full coefficients. Keeping those two outputs separate prevents a lower-bound calculation from being mistaken for an exact coefficient evaluation.

13.4 Reproduction commands and dependencies

The article is self-contained LaTeX and uses standard packages. From the package directory, run

```
python code/verify.py -order 12
python code/diagnostics.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The supplied `build.sh` runs the three document passes. The exact script needs no third-party Python dependency; only the diagnostic script needs NumPy and SciPy. The scripts make no network requests. The package contains the scripts, their recorded outputs, source notes, and build instructions. No source or branch in ProveIt was modified.

14 Formalization targets and proof dependencies

The repository already has formal infrastructure for asymptotic scales, flatness, well-based supports, and differential blocks [1]. Those declarations do not automatically formalize this article’s infinite analytic constructions. A useful implementation should separate four layers and record the exact theorem supplied by each.

14.1 Finite coefficient algebra

The first layer should define weighted compositions and the finite Lagrange coefficients over a commutative algebra, prove the recurrence (13.1), and identify the signed inverse blocks. Rational specializations can then produce kernel-checked finite identities. Positivity and the one-action injection give a particularly small initial target. These lemmas have no dependence on Borel summation, asymptotic estimates, or numerical special functions.

14.2 Real estimates and the factorial budget

The next layer should establish the block mass estimate, the cost-moment bound, factorial inequalities, and the uniform exponential Taylor remainder. The compact statement of Lemma 6.1 would be a reusable library result. A theorem about a family of finite moments must be distinguished from a theorem about a function remainder; the geometric tail and uniform-limit arguments connect them.

14.3 Analytic realization and summation

The analytic layer needs uniform convergence on closed half-discs, normal convergence in open domains, the fixed-parameter inverse theorem, and the beta-measure kernel identity. The two Laplace exchanges have different hypotheses: absolute Taylor integration for one frequency, and an oscillatory absolute-integral bound for infinitely many blocks. Formalizing them as separate lemmas would guard against a common invalid exchange. A general fine-summability algebra should then package the convolution and inverse theorem in Appendix A.

14.4 Sharp asymptotics and probability

The final layer contains the entropy estimate, the two length ranges in Lemma 8.1, and the conversion of exponential losses into probability concentration. The statement must distinguish a limsup type, an all-degree root limit, and a multiplicative equivalent. The current argument establishes the first two in their respective hypotheses, not the third. It should likewise distinguish concentration of the largest cost from any unproved local central limit theorem. No Lean files are supplied or claimed to have been checked in this package; this section is an implementation roadmap.

15 Further research questions

The following are proposed next tasks for this weighted model. They are not blanket assertions that every analogous question is open throughout the literature; comparison with general type-preservation and summation theorems should precede a publication-priority claim.

15.1 Sharp signed inverse coefficients

Determine the exact type and, more ambitiously, signed coefficient asymptotics for $\hat{Q}$ under strong damping. The parabolic construction already gives $T_1(\hat{Q}) \leq T_1(\hat{U})$ in the finite positive-type case. A general exact-type-preservation theorem, if applicable with this normalization, should be used rather than rediscovered. Even such a theorem would not settle the signs, cancellation scale, or multiplicative prefactor of the inverse coefficients.

15.2 A full multiplicative forward asymptotic

For $d_j = bj^\beta$ and $\lambda_j = aj^p$, refine (9.1) to a multiplicative equivalent. The dominant large action has size $n^{1/\beta}$, but small nonunit parts, the cost of using fewer than n parts, and lattice rounding all contribute below order n . Determine how many small actions must be retained to obtain relative error $o(1)$, and whether a finite-core threshold depends on both p and β rather than on their ratio alone.

15.3 Local fluctuations and a conditional ensemble

Prove a local limit theorem for $J_{\max}$ and a fluctuation theorem for K under (9.2). A formal quadratic expansion of the cost profile suggests a first scale, but it does not account for the constraint that total action is exactly n . The first objective is a correct centering, including the drift from the sea of small actions; a Gaussian formula with the wrong center is not a local limit theorem.

15.4 Lacunary cost spectra and log-periodic behavior

For geometrically spaced costs, such as $d_j = e^{\theta j}$, the type proof obtains a sharp limsup by choosing degrees matched to individual costs. Determine the lower root envelope at unmatched degrees, whether several expensive actions can compensate for a gap, and whether the normalized logarithms have periodic limit profiles in $\log n$. The dense power-cost theorem does not decide this question.

15.5 Actual inverse Borel singularities

In the finite positive-type regime, determine the maximal continuation domain of B_Q , beginning with the positive intercept of (10.1). Does the first forward singularity survive reversion at the same Borel location? Determine its local form and any lateral summation discontinuity in specific power-cost models. Normal convergence of a chosen kernel representation gives only one side of this problem.

15.6 Angular and generalized moment summation

Strengthen fine negative-ray summability to an open arc of directions where possible, and identify obstructions where it fails. For $s \neq 1$, construct moment kernels appropriate to $(n!)^s$ or $\Gamma(1 + sn)$, keeping the exact conversion of types visible. The target should identify the sum with the literal

inverse, not merely prove the existence of some function with the same asymptotic expansion. Uniform Gevrey remainders are a starting point, not the entire summation argument.

15.7 Amplitudes beyond exponential growth

The counterexample (12.1) shows that slope control alone cannot handle rapidly growing positive amplitudes. Seek a joint criterion in terms of the amplitude generating function’s Borel continuation and the cost of its interaction with the slopes. Even the zero-slope model forces directional singularity information into such a criterion. This suggests separating an amplitude summation class from a feedback stability condition.

15.8 Complex phases and vector feedback

For signed amplitudes or complex slopes, identify usable noncancellation or sectorial-damping conditions. For a vector equation, replace the scalar exponential feedback by linear forms in several unknowns and ask whether a matrix or convex-geometric budget controls regularity. A countable-tail condition must be explicit; a finite-dimensional implicit-function theorem cannot control an unbounded family of coefficients by itself.

15.9 General actions and missing action one

Replace q^j by q^{a_j} for integer or well-based real actions. Determine which version of the budget survives after replacing the index j by action size. The present lower bound uses arbitrarily many action-one factors and the present inverse uses an ordinary nonzero linear coefficient. When the smallest action is different, ramification or a different formal inverse object must be specified before an analytic statement can be formulated.

15.10 Certified algorithms with adaptive costs

The universal factor 8^k is simple but wasteful. Develop certified weighted mass generating functions and choose adaptively between exact block truncation, Taylor truncation, and Borel quadrature. The amplitude cost is available before a frequency is evaluated and can guide pruning of negligible compositions. A useful deliverable would combine a formally checked finite certificate with an explicit analytic tail bound, rather than rely on floating cancellation at high order.

16 Conclusion

The positive weighted feedback model admits a sharp and explicit answer to the amplitude–slope question. Ordinary analyticity is governed by $j + d_j$, while every positive Gevrey order is governed by $j \log(j + 1) + d_j$. The same latter budget controls an actual uniform left-side inverse remainder; at order one it also characterizes fine negative-direction summability and identifies the summed equation.

Strong damping produces a sharper theorem: an exact type controlled by λ_j/d_j^{s+1} , and, for dense power costs, an all-degree root limit and concentration of one dominant action. These are model-specific results with complete arguments and reproducible finite checks. They do not rely on accepting an unreviewed repository draft as a proof, and they leave the multiplicative asymptotics, signed inverse behavior, maximal Borel continuation, and stronger summation questions explicitly separate.

A Compositional inversion in the fine-summable class

This appendix proves the nonlinear result used in Theorem 2.4. It is included because a theorem stated for an open angular sector cannot silently be applied to a fixed-width tube. The proof specializes the classical convolution method described in [5].

A.1 The Borel minor at infinity

For $\varphi(z) = \sum_{n \geq 1} \varphi_n z^{-n}$ define

$$\mathcal{B}\varphi(\xi) = \sum_{n \geq 1} \frac{\varphi_n}{(n-1)!} \xi^{n-1}.$$

Let $\Omega_\delta^+ = -\Omega_\delta^-$, which is convex. Suppose a minor $\widehat{\varphi}$ is holomorphic there and $|\widehat{\varphi}(\xi)| \leq M e^{H|\xi|}$. Products of formal series correspond to convolution along the segment from zero to ξ . The convolution simplex has volume $1/(k-1)!$, and convexity keeps all its arguments in the tube. Thus

$$|\widehat{\varphi}^{*k}(\xi)| \leq M^k \frac{|\xi|^{k-1}}{(k-1)!} e^{H|\xi|}. \quad (\text{A.1})$$

The sum of the lengths of the collinear arguments is $|\xi|$, which explains the single exponential on the right.

If $F(w) = \sum F_k w^k$ is convergent near zero, with $|F_k| \leq C R^{-k}$, (A.1) proves normal convergence of the minor of $F(\varphi) - F(0)$ and an exponential bound. In particular, reciprocals of formal series with nonzero constant term are permitted. Multiplication by z , after removal of the new constant term, is differentiation of the minor and preserves its exponential bound on a narrower tube by Cauchy's estimate. Division by z is integration. Translation $z \mapsto z + c$ multiplies the minor by $e^{-c\xi}$. All these operations preserve this fine-summable class.

Laplace summation

$$\mathcal{S}\varphi(z) = \int_0^\infty e^{-z\xi} \widehat{\varphi}(\xi) d\xi, \quad \operatorname{Re} z > H,$$

commutes with the stated operations. For products this is the absolutely convergent convolution integral and Fubini's theorem; for analytic scalar substitution it follows from the same exponential majorant. Multiplication by z follows by integration by parts, and translation follows directly from its multiplier. These arguments also give the estimates $|\mathcal{S}\varphi(z)| \leq M/(\operatorname{Re} z - H)$ and $|\partial_z \mathcal{S}\varphi(z)| \leq M/(\operatorname{Re} z - H)^2$.

A.2 An explicit inverse minor

Lemma A.1 (Fine inversion at infinity). *Let $f(z) = z + c + \varphi(z)$, where $\varphi = O(z^{-1})$ has a minor holomorphic with an exponential bound on some Ω_δ^+ . Its formal inverse has the form $g(z) = z - c + \chi(z)$ with $\chi = O(z^{-1})$ in the same class, possibly on a narrower tube. Their Laplace sums are analytic inverses on sufficiently far right half-planes and their corresponding images.*

Proof. First take $c = 0$. Formal Lagrange inversion gives

$$\chi(z) = \sum_{k \geq 1} \frac{(-1)^k}{k!} \partial_z^{k-1} (\varphi(z)^k). \quad (\text{A.2})$$

For example, apply coefficient Lagrange inversion to $w = -t\varphi(z + w)$ and then put $t = 1$; the z^{-1} -order of the k th summand tends to infinity. Differentiation corresponds to multiplication of a Borel minor by $-\xi$, so the proposed inverse minor is

$$\widehat{\chi}(\xi) = - \sum_{k \geq 1} \frac{\xi^{k-1}}{k!} \widehat{\varphi}^{*k}(\xi). \quad (\text{A.3})$$

Equation (A.1) bounds its absolute value by

$$e^{H|\xi|} \sum_{k \geq 1} \frac{M^k |\xi|^{2k-2}}{k!(k-1)!} \leq M e^{(H+2\sqrt{M})|\xi|}.$$

The last inequality follows, for $t \geq 0$, from $\sum_{\ell \geq 0} t^{2\ell} / (\ell!(\ell+1)!) \leq e^{2t}$, using $\binom{2\ell}{\ell} \leq 4^\ell$. This also proves normal convergence and identifies the formal coefficients.

It remains to identify the analytic sum, rather than just its formal inverse coefficients. On a far right half-plane, the Laplace sum $\phi(z) = \mathcal{S}\varphi(z)$ is uniformly small and has uniformly small derivative. The equation $w = -t\phi(z + w)$ has a holomorphic solution for $|t|$ in a disc of radius greater than one: choose a fixed small disc for w and move the half-plane sufficiently far right to make the map a strict contraction there. Classical analytic Lagrange inversion yields (A.2) with ϕ in place of φ , evaluated at $t = 1$.

The Laplace integral of the absolute k th minor is at most

$$\frac{M^k (2k-2)!}{k!(k-1)!(\operatorname{Re} z - H)^{2k-1}},$$

which is summable when $\operatorname{Re} z - H > 2\sqrt{M}$. Thus summation of (A.3) commutes with its Laplace integral and gives exactly that analytic Lagrange series. The sum $z + \mathcal{S}\chi(z)$ is the analytic inverse. Injectivity on a far right half-plane also follows by integrating the derivative bound $|\phi'| < 1/2$ along line segments in that convex domain. For general c , invert $f_0(z) = z + \varphi(z)$ first and then translate: $f^{-1}(z) = f_0^{-1}(z - c)$. Translation preserves all the established properties. $\square$

A.3 Returning to series tangent to the identity at zero

Let $Q(u) = u + O(u^2)$ be a fine-summable formal series in direction π . Conjugate it by $T(z) = -1/z$:

$$F(z) = T \circ Q \circ T(z) = -\frac{1}{Q(-1/z)} = z + c + \varphi(z).$$

Changing the sign of the Borel variable moves the negative tube to the positive one. The minor at infinity differs from the integrated Borel transform only by differentiation, and the reciprocal and multiplication operations used to form F are permitted by the preceding algebra. Lemma A.1 constructs $G = F^{-1}$ in the same class and identifies its sum. Conjugating back gives

$$T \circ G \circ T = Q^{\langle -1 \rangle},$$

which is finely summable in direction π . Summation commutes with the conjugation operations, so the sums are inverse functions on their small common domains. This is the exact closure result used in the main text; no angular-sector hypothesis has been inserted.

B Notation and the logical dependency map

Symbol	Meaning
c_j, d_j	Positive amplitude and its cost $-\log c_j$
λ_j	Nonnegative feedback slope at action j
u_n, v_n	Forward and inverse ordinary coefficients
L_j, H_j	$j \log(j+1)$ and $j \log(j+1) + d_j$
s, r	Gevrey order $s > 0$ and $r = s + 1$
T_s	Type using the normalization $(n!)^s$
k, D, b	Inverse block number, composition cost, frequency
$K, D, J_{\max}$	Lagrange-ensemble length, cost, largest action
$\mathcal{B}_0, \mathcal{B}$	Integrated Borel transform at zero; minor at infinity
$h, \tilde{h}$	Absolute real and imaginary parts of $\sqrt{\zeta}$

The repeated letter D always denotes a sum of amplitude costs; its indexing composition is clear from the context. The inverse block index k is not the length ℓ of its internal composition. The Lagrange length K belongs instead to the forward probability ensemble.

The proof dependencies can be summarized without identifying unlike statements. Finite Lagrange algebra gives positivity and the necessary budget. Weighted inverse blocks, the moment estimate, and the factorial lemma give sufficient Gevrey bounds and uniform inverse remainders. Bessel kernels plus the weighted tube estimate give fine inverse summation, and the nonlinear lemma transfers it to the forward series. Separately, a positive variational upper bound and a one-action lower bound give the exact strong-damping type. Dense power costs make the lower bound available at every degree and then turn strict variational losses into probability concentration. None of these arrows uses the finite computational checks as a substitute for an infinite proof.

References

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